CELEBAL EXCELLENCE INTERNSHIP PROGRAM

DATA ENGINEERING

WEEK - 7 ASSIGNMENT REPORT

DATABRICKS

SUBMITTED BY:-

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(JKLU)

1. OBJECTIVE

The objective of this assignment was to implement incremental data processing using Delta Lake. The pipeline demonstrates how raw data is ingested, cleaned, updated using incremental records, and transformed into analytics-ready tables using Apache Spark and Delta Lake.

The main focus of this implementation is the Delta Lake MERGE operation, which enables efficient updating of existing records and inserting new records without rewriting the complete dataset.

2. TECHNOLOGIES USED

- Databricks
- Apache Spark
- PySpark
- Delta Lake

3. DATA PIPELINE ARCHITECTURE

The implementation follows the Medallion Architecture approach:

Raw CSV Dataset → Bronze Layer → Silver Layer → Incremental MERGE Processing → Gold Layer

4. BRONZE LAYER - Raw Data Ingestion

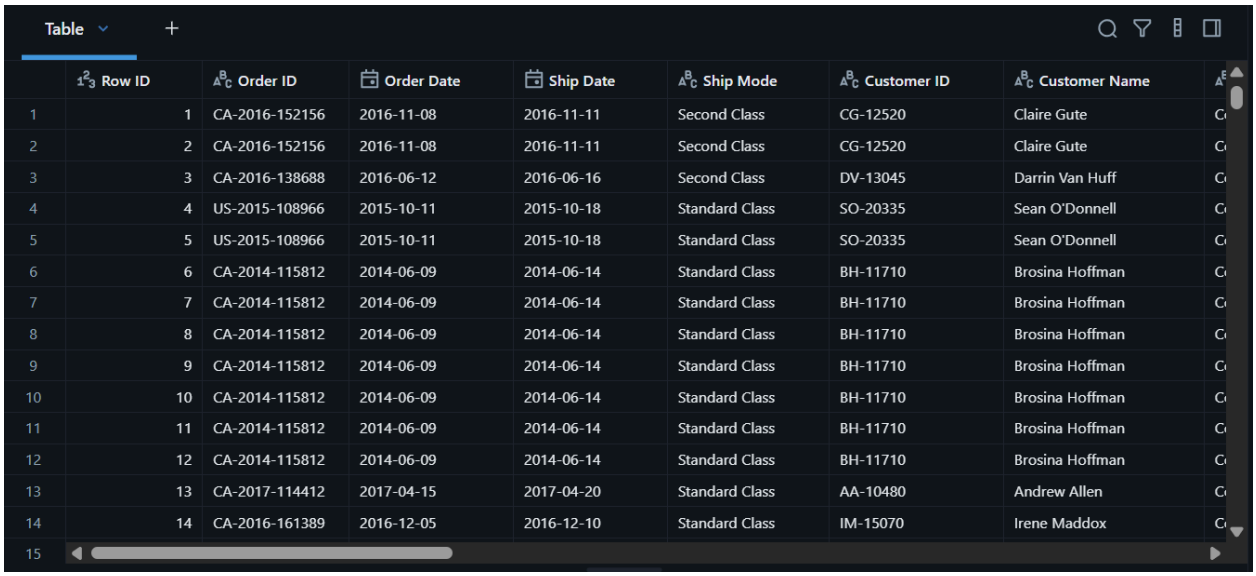
The Sample Superstore dataset was loaded from the Unity Catalog Volume into a Spark DataFrame.

The raw dataset was stored as a Delta table named: **“bronze”**

The Bronze layer stores the data in its original form and acts as the initial storage layer before performing any transformations.

Operations performed:

- Loaded CSV file using Spark DataFrame API
- Enabled header reading and schema inference
- Stored the dataset in Delta format



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	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	C
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	C
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	C
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	C
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	C
6	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
7	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
8	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
9	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
10	10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
11	11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
12	12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	C
13	13	CA-2017-114412	2017-04-15	2017-04-20	Standard Class	AA-10480	Andrew Allen	C
14	14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	Irene Maddox	C
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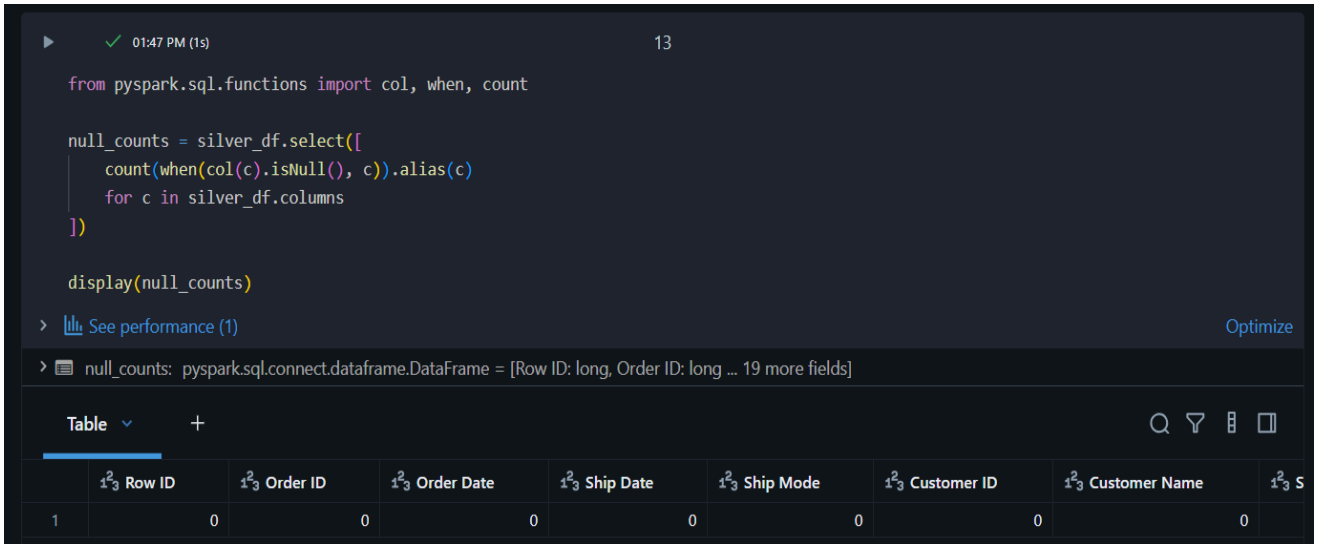
5. SILVER LAYER - Data Cleaning

The Bronze data was processed to create a cleaned Silver layer.

The following cleaning operations were performed:

Null Value Checking

All columns were checked for missing values.

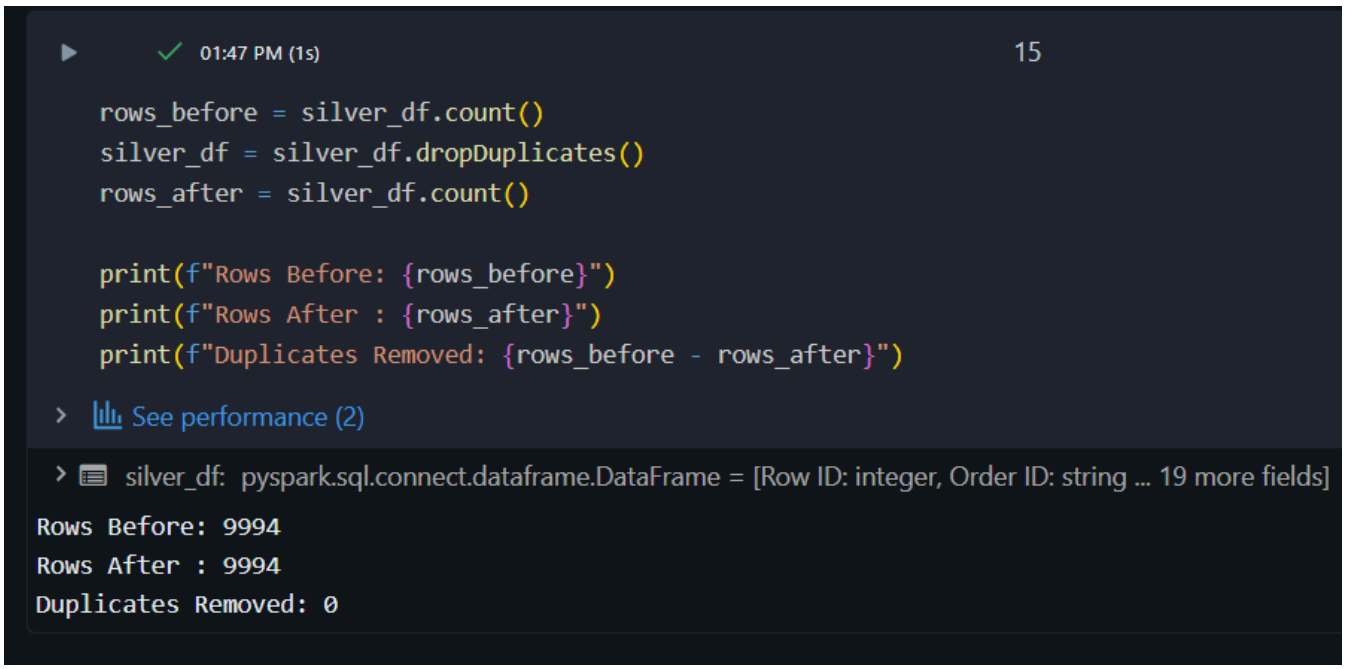


Observation:

- No null values were found in the dataset.
- Therefore, no additional null handling was required.

Duplicate Removal

Duplicate records were checked and removed using: **dropDuplicates()**



Observation:

- No duplicate records were found.
- The row count remained unchanged.

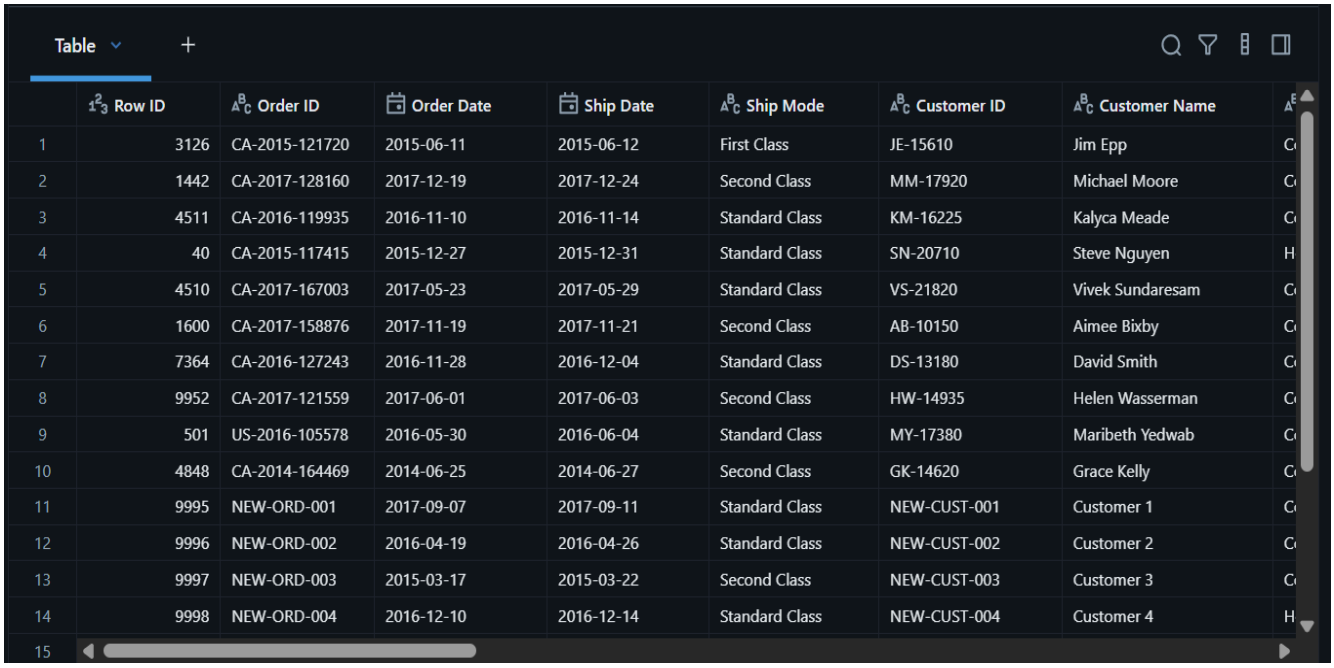
The cleaned dataset was stored as a Delta table: **“silver”**

6. INCREMENTAL DATA LOADING

A second dataset was introduced to simulate new incoming data.

The incremental dataset **“customer_incremental.csv”** was loaded and stored as a Delta table **“incremental”**

This dataset contained records that needed to be merged into the existing Silver table.



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	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	
1	3126	CA-2015-121720	2015-06-11	2015-06-12	First Class	JE-15610	Jim Epp	C
2	1442	CA-2017-128160	2017-12-19	2017-12-24	Second Class	MM-17920	Michael Moore	C
3	4511	CA-2016-119935	2016-11-10	2016-11-14	Standard Class	KM-16225	Kalyca Meade	C
4	40	CA-2015-117415	2015-12-27	2015-12-31	Standard Class	SN-20710	Steve Nguyen	H
5	4510	CA-2017-167003	2017-05-23	2017-05-29	Standard Class	VS-21820	Vivek Sundaresam	C
6	1600	CA-2017-158876	2017-11-19	2017-11-21	Second Class	AB-10150	Aimee Bixby	C
7	7364	CA-2016-127243	2016-11-28	2016-12-04	Standard Class	DS-13180	David Smith	C
8	9952	CA-2017-121559	2017-06-01	2017-06-03	Second Class	HW-14935	Helen Wasserman	C
9	501	US-2016-105578	2016-05-30	2016-06-04	Standard Class	MY-17380	Maribeth Yedwab	C
10	4848	CA-2014-164469	2014-06-25	2014-06-27	Second Class	GK-14620	Grace Kelly	C
11	9995	NEW-ORD-001	2017-09-07	2017-09-11	Standard Class	NEW-CUST-001	Customer 1	C
12	9996	NEW-ORD-002	2016-04-19	2016-04-26	Standard Class	NEW-CUST-002	Customer 2	C
13	9997	NEW-ORD-003	2015-03-17	2015-03-22	Second Class	NEW-CUST-003	Customer 3	C
14	9998	NEW-ORD-004	2016-12-10	2016-12-14	Standard Class	NEW-CUST-004	Customer 4	H
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7. Delta Lake MERGE Operation

The main objective of this assignment was achieved using the Delta Lake MERGE operation.

The MERGE operation performs an upsert process:

- If a matching record exists, the existing record is updated.
- If a matching record does not exist, a new record is inserted.

The merge condition was based on **"Row ID"**

The implementation used **"whenMatchedUpdateAll()"** to update existing records.

And **"whenNotMatchedInsertAll()"** to insert new records.

This allows incremental changes to be processed efficiently without replacing the complete dataset.

MERGE Result

Before MERGE: Silver Table Records: 9994, Incremental Records: 20

After MERGE: Final Silver Table Records: 10004

Based on the final row count, the MERGE operation successfully updated existing records and inserted new records into the Silver Delta table without creating duplicate Row IDs.

8. VALIDATION

After performing the MERGE operation, validation checks were performed.

The following validations were completed:

Row Count Validation- Final record count was checked to confirm successful processing.

Result: Final Row Count: 10004

Duplicate Validation- Duplicate Row IDs were checked after MERGE.

Result: Duplicate Row IDs: 0

This confirms that the MERGE operation maintained data consistency.

9. GOLD LAYER - Analytical Tables

After completing incremental processing, business-ready tables were created from the updated Silver layer.

Fact Table

The fact table contains transactional information such as:

- Order details
- Sales
- Quantity
- Discount
- Profit

Created table: **"fact_table"**

Customer Dimension

The customer dimension stores customer-related attributes:

- Customer ID

- Customer Name
- Segment
- Location details

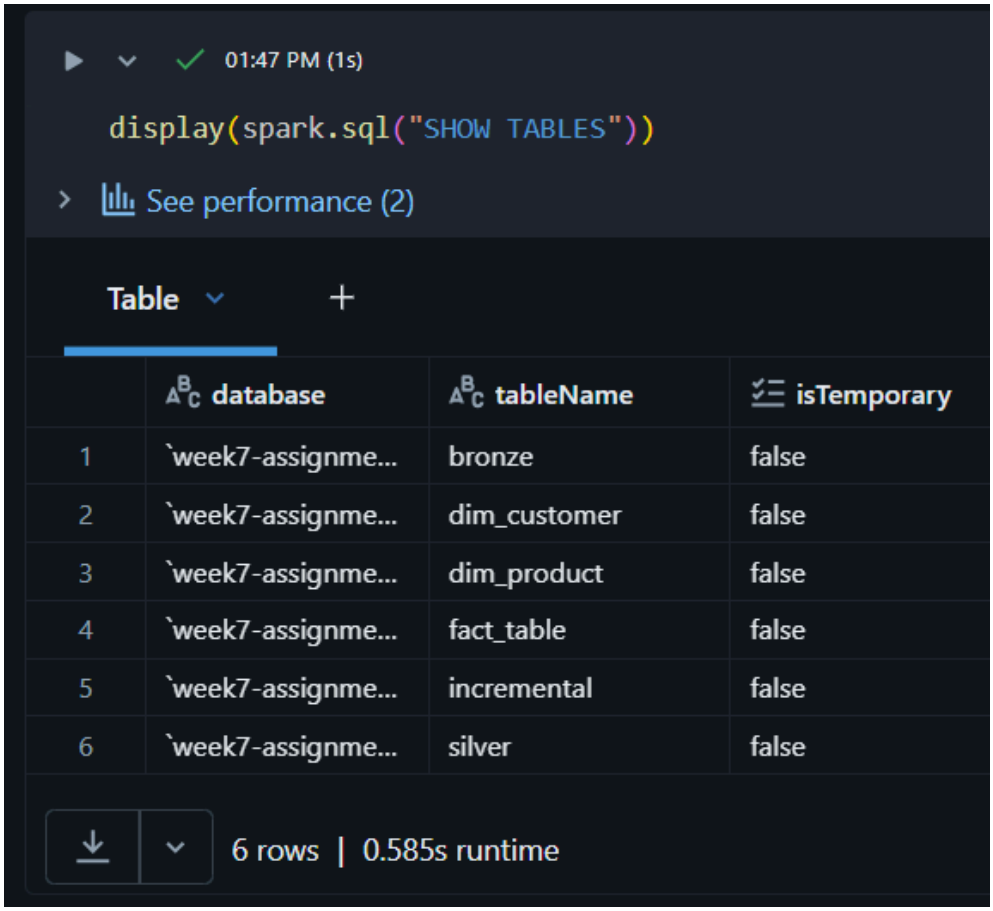
Created table: **“dim_customer”**

Product Dimension

The product dimension stores product-related attributes:

- Product ID
- Product Name
- Category
- Sub-category

Created table: **“dim_product”**



10.CONCLUSION

This assignment successfully demonstrated incremental data processing using Delta Lake in Databricks. The Medallion Architecture was used to organize data into Bronze, Silver, and Gold layers. Delta Lake's MERGE operation efficiently handled updates and inserts from an incremental dataset while maintaining data consistency. Finally, analytical fact and dimension tables were created to support downstream reporting and analysis.